

UNIT 2 – GAUSS'S LAW

Unit 2 Review**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

1. A thin rod of length L carries total charge Q spread uniformly along it. Find the electric field at a point on the rod's axis a distance a from the nearer end.

Let $\lambda = Q/L$ and put the near end at $x = 0$ with the field point at $x = -a$. A slice $dq = \lambda dx$ at position x is a distance $a+x$ away, so $dE = \frac{k\lambda dx}{(a+x)^2}$. Integrating: $E = k\lambda \int_0^L \frac{dx}{(a+x)^2} = k\lambda \left[\frac{1}{a} - \frac{1}{a+L} \right] = \frac{kQ}{a(a+L)}$, pointing away from the rod.

2. A closed surface encloses point charges $q_1 = +2.0$ nC and $q_2 = -5.0$ nC (and a third charge of $+7.0$ nC sits outside). Find the net electric flux through the surface.

Only enclosed charge counts: $\Phi_E = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{-3.0 \times 10^{-9}}{8.85 \times 10^{-12}} = -3.4 \times 10^2 \text{ N} \cdot \text{m}^2/\text{C}$.

3. An infinitely long line of charge has linear density $\lambda = 2.0 \mu\text{C}/\text{m}$. Find E at a perpendicular distance of 0.050 m.

Gaussian cylinder of radius r and length ℓ : $E(2\pi r\ell) = \lambda\ell/\epsilon_0$, so $E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r} = \frac{2(8.99 \times 10^9)(2.0 \times 10^{-6})}{0.050} = 7.2 \times 10^5 \text{ N/C}$.